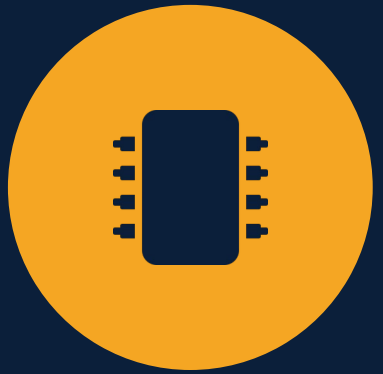
UNIT 3 · 9 hours

Full-Wave Bridge Rectifier



Diodes, Transistors and their Applications

Basic Electrical and Electronics Engineering (EEE103)
Dr. Mitesh Astik · Dept. of Electrical and Electronics Engineering
NMAM Institute of Technology, Nitte · 2025-26

U3

Learning objectives

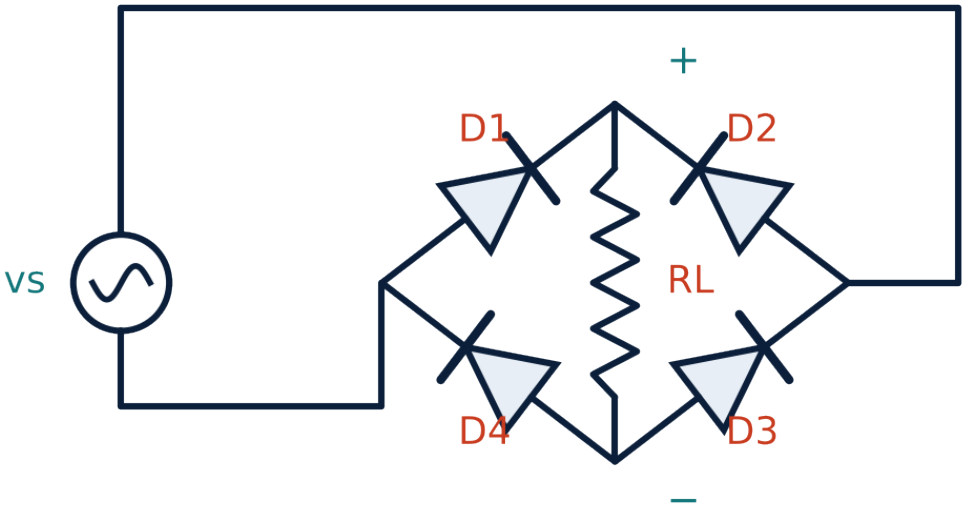
- Explain the operation of a bridge rectifier
- Derive and use Vdc, Idc, Irms, ripple factor and efficiency
- Compare half-wave, centre-tap and bridge rectifiers
- Solve numericals

Syllabus context

Semiconductor diode, diode equivalent circuits, half-wave rectifier, full-wave bridge rectifier. Zener diode and its use in voltage regulation.

U3

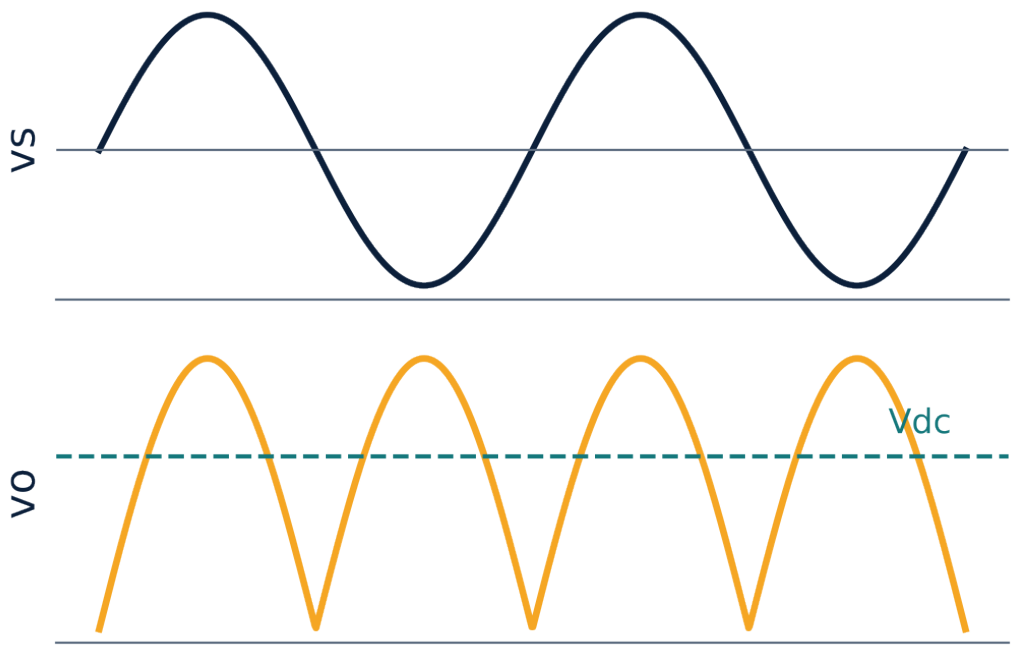
Bridge rectifier circuit



- Four diodes D1–D4 in a bridge.
- Positive half-cycle (top terminal +): D2 and D4 conduct; current through RL top → bottom.
- Negative half-cycle: D1 and D3 conduct; current through RL again top → bottom.
- No centre-tapped transformer needed.

U3

Input and output waveforms



- Both half-cycles reach the load.
- vo = |Vm sin θ|; output frequency = 2 f.
- Vdc = 2Vm/π (dashed line).

U3

Performance equations

$$I_m = V_m / (2R_f + R_L)$$

two diodes in path

$$I_{dc} = 2 I_m / \pi$$

average current

$$I_{rms} = I_m / \sqrt{2}$$

RMS current

$$V_{dc} = 2 V_m / \pi$$

DC output (Rf = 0)

U3

Efficiency, ripple and PIV

$\eta = 81.2\ \%$	$\gamma = 0.482$	$PIV = V_m$
maximum (Rf = 0)	*ripple factor*	*each diode (bridge)*

- Centre-tap full-wave needs PIV = 2Vm and a centre-tapped transformer.
- Bridge uses the transformer secondary fully: TUF ≈ 0.81 compared with 0.287 (half-wave).

U3

Rectifier comparison

Parameter	Half-wave	Centre-tap FW	Bridge FW
Diodes	1	2	4
Vdc	Vm/π	2Vm/π	2Vm/π
Ripple factor	1.21	0.482	0.482
Efficiency (max)	40.6 %	81.2 %	81.2 %
PIV	Vm	2Vm	Vm
Output frequency	f	2f	2f

U3

Numerical: bridge rectifier

Problem A bridge rectifier is fed from 240 V RMS, RL = 480 Ω, each diode Rf = 2 Ω. Find mean load current and power in each diode.

- $V_m = 240\sqrt{2} = 339.4$ V; $I_m = 339.4/(480 + 2 \times 2) = 0.701$ A
- $I_{dc} = 2I_m/\pi = 0.446$ A
- Each diode conducts on one half-cycle: $I_{(rms, diode)} = I_m/2 = 0.351$ A
- $P_{diode} = (I_m/2)^2 R_f = 0.351^2 \times 2$

Answer: $I_{dc} = 0.446$ A, $P_{(diode)} \approx 0.25$ W

U3

Key takeaways

- Bridge: four diodes, two conduct in each half-cycle
- Vdc = 2Vm/π, Idc = 2Im/π, ripple 0.482, η max 81.2 %
- PIV = Vm for the bridge
- No centre tap required

U3

Practice questions

Q1. Explain the operation of a bridge rectifier with a diagram.

Q2. Compare half-wave and full-wave rectifiers.

Q3. A bridge circuit gives 12 V DC (Rf = 0). Find the RMS secondary voltage.

Q4. Why does the full-wave rectifier have lower ripple?